CAPSTONE PROJECT: EXPLORING COFFEE QUALITY DATA WITH POWER BI

By
Soumyo Bhattacherjee
(S12160)

TABLE OF CONTENT

PROJECT OVERVIEW

The goal of this project is to analyze the CQI dataset to uncover the key sensory attributes—such as aroma, flavor, and acidity—that determine coffee quality, examine correlations between processing methods, origin regions, and overall quality scores, identify patterns in defect occurrences and their impact on quality outcomes, investigate.

Also, this project will deprecate multiple factors interact and influence the Total Cup Points, which serve as a comprehensive measure of coffee quality, and provide actionable insights that can guide producers, traders, and stakeholders in improving coffee quality and consistency.

COFFEE QUALITY INSTITUTE

The Coffee Quality Institute (CQI) is a non-profit organization that works to improve the quality and value of coffee worldwide. It was founded in 1996 and has its headquarters in California, USA.

CQI's mission is to promote coffee quality through a range of activities that include research, training, and certification programs. The organization works with coffee growers, processors, roasters, and other stakeholders to improve coffee quality standards, promote sustainability, and support the development of the specialty coffee industry.

TECHNOLOGY TOOLKIT

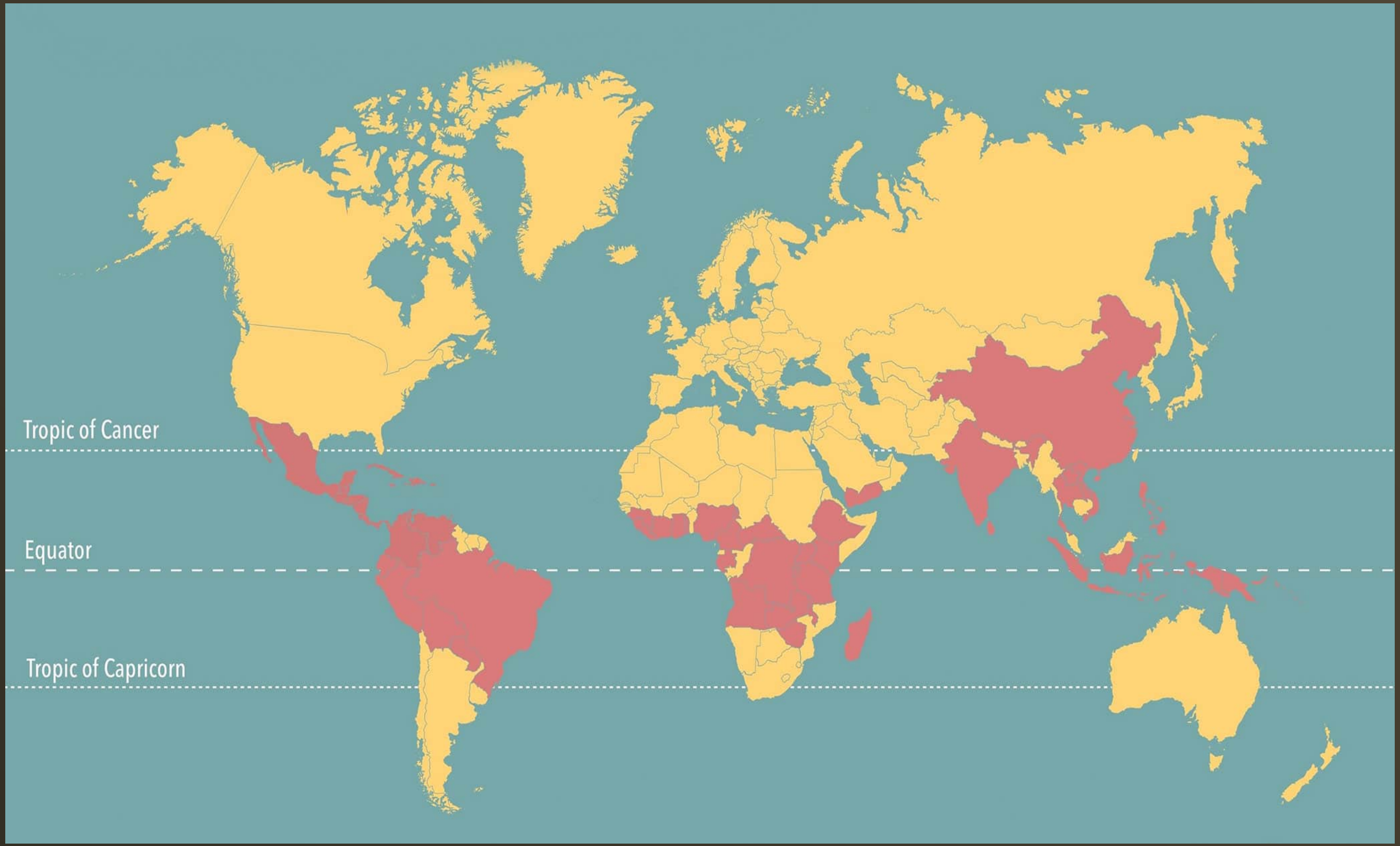
Power BI

Excel

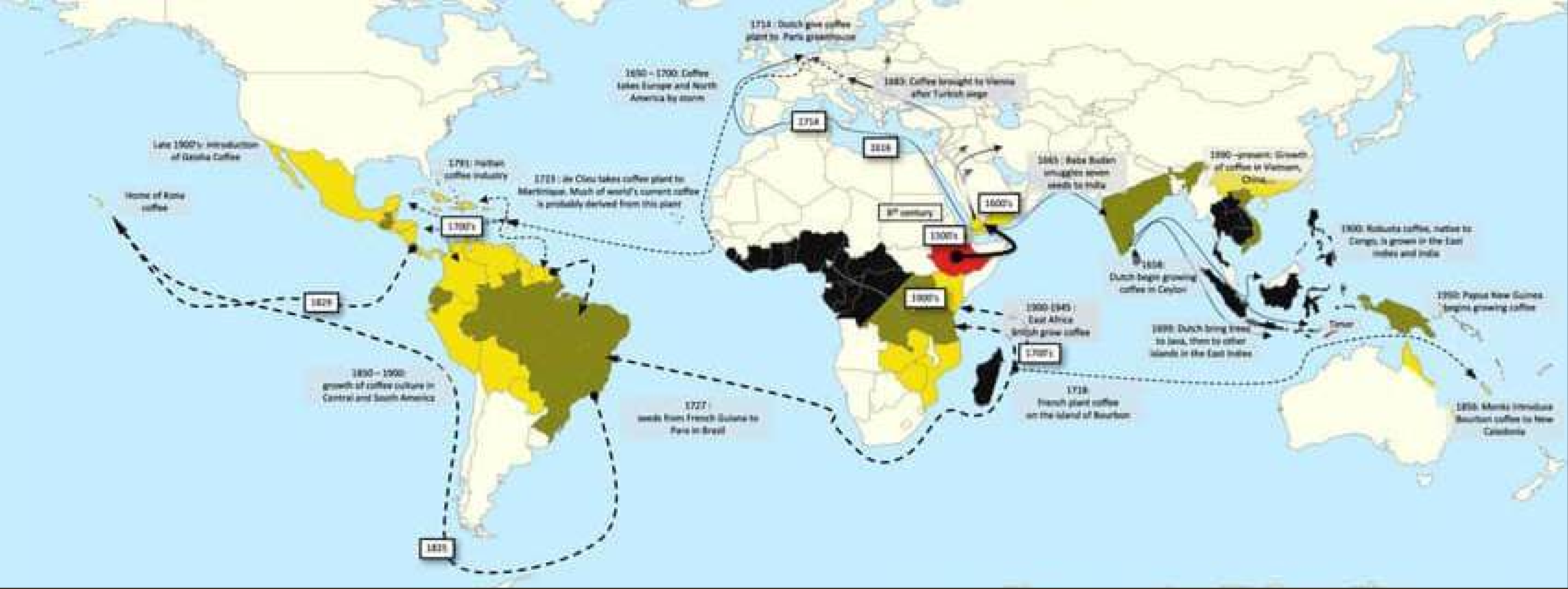
PowerPoint

COFFEE PRODUCING REGION AS PER DATASET

ABOUT COFFEE PRODUCTION



Coffee is produced in countries within the equatorial "bean belt," with major producers including Brazil, Vietnam, and Colombia. The process starts with planting seedlings in nurseries, which grow into trees that flower and produce fruit with coffee beans. Once harvested, these beans are processed through methods like the washed or natural process to remove the fruit pulp before drying, grading, and bagging



The data includes a range of information on coffee production, processing, and sensory evaluation. It also contains data on coffee genetics, soil types, and other factors that can affect coffee quality.

DATA

SENSORY EVALUATIONS (COFFEE QUALITY SCORES)

Aroma – This refers to the fragrance of coffee, which is first experienced when the beans are ground and brewed. Aroma is a crucial aspect of coffee quality as it sets the stage for the overall sensory experience. Notes can range from floral and fruity to nutty, spicy, or chocolate-like.

Flavor – Flavor is the combined impression of taste and aroma once the coffee is in the mouth. It includes sweetness, bitterness, acidity, and distinctive notes such as fruity, nutty, or chocolate undertones. Flavor is considered one of the most important quality indicators.

Aftertaste – Also known as the “finish,” this is the taste that lingers in the mouth after swallowing. A pleasant aftertaste is smooth, clean, and long-lasting, whereas an undesirable one may be harsh or leave bitter or sour notes.

Acidity – In coffee, acidity does not mean sourness. Instead, it refers to brightness, liveliness, or sparkle on the palate. High-quality acidity is crisp and refreshing, often described with citrus or berry-like notes.

Attribute	Score
Fragrance	8
Flavor	8
Aftertaste	7.5
Acidity	8.25
Body	8
Balance	8
Uniformity	10
Clean Cup	10
Sweetness	10
Overall	8
Total score	85.75
Defect	0
Final score	85.75

SENSORY EVALUATIONS (COFFEE QUALITY SCORES) – CONTD…

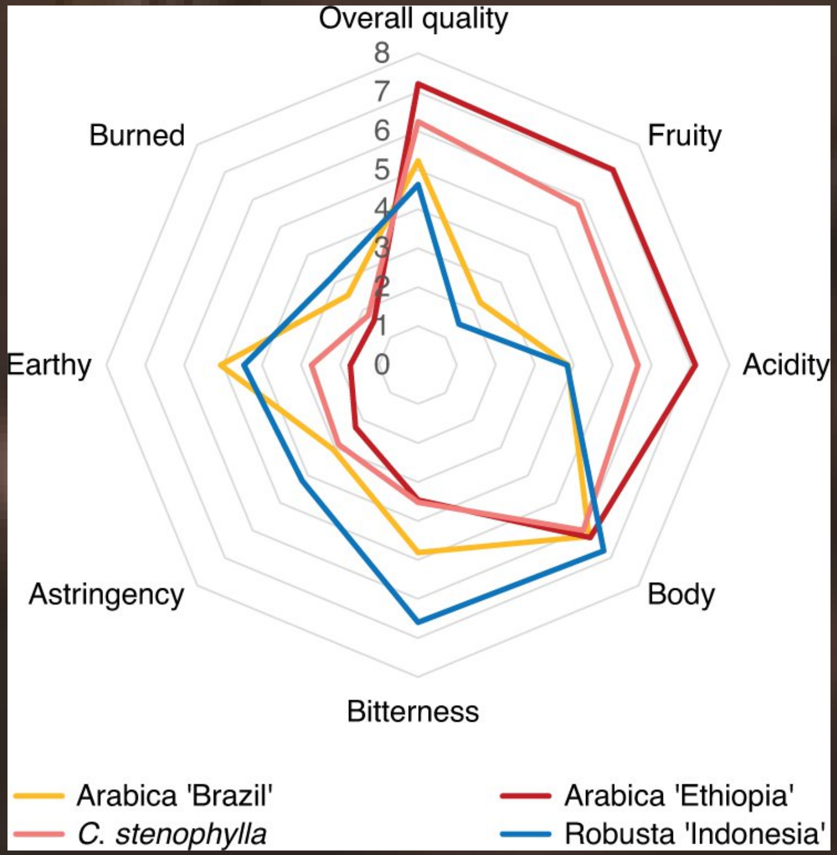
Body – Body describes the weight, texture, or mouthfeel of coffee. A coffee with “full body” feels thick, creamy, or heavy on the tongue, while a “light body” feels thin and tea-like.

Balance – Balance reflects how well the coffee’s different components (flavor, acidity, body, aftertaste, sweetness) complement each other. A well-balanced coffee has no single attribute overwhelming the rest.

Uniformity – This measures how consistent the coffee tastes across multiple cups brewed from the same batch. High uniformity indicates reliable quality with no variations or irregular flavors between cups.

Clean Cup – A clean cup means the coffee is free of off-flavors, taints, or defects such as sourness, mustiness, staleness, or earthy notes that don’t belong. It represents purity and clarity of flavor.

Sweetness – Sweetness is a highly desirable trait in coffee, often described as caramel-like, honeyed, fruity, or floral. It balances acidity and bitterness, contributing to a pleasant and smooth flavor profile.



DATA PRE PROCESSING IN EXCEL AND POWER QUERY EDITOR

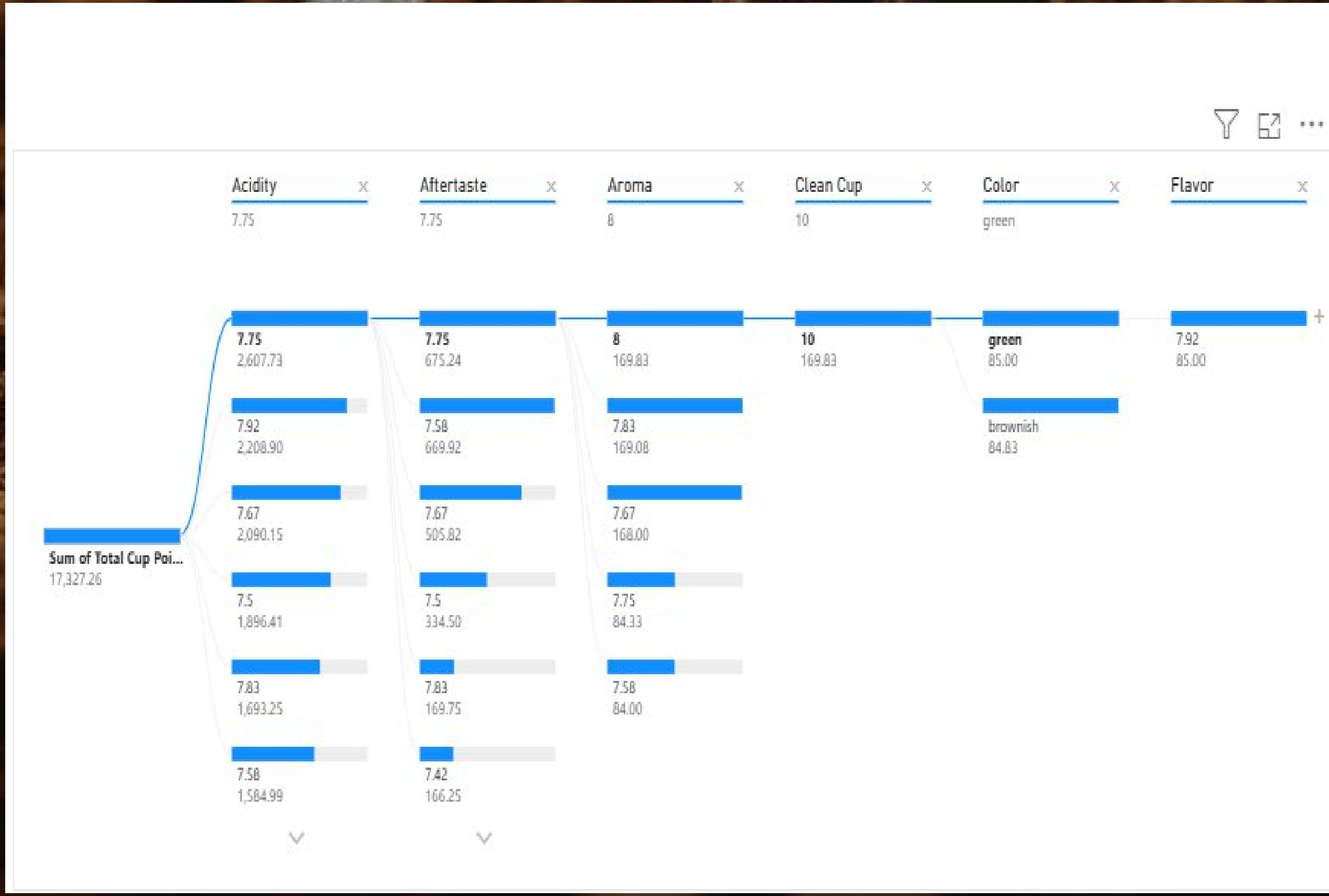
- **ID Uniqueness Check**
  The dataset is examined to verify whether all IDs are unique.
- **Column Translation**
  The values in the **Lot Number** and **Region** columns are translated into English using Google Translate.
- **Altitude Handling**
  Where altitude is provided as a range (e.g., 1200–1400), the mean value is calculated and used to replace the range.
- **Harvest Start Year Extraction**
  A new column, **Harvest Start Year**, is created by extracting the starting year from the **Harvest Year** column.
- **Missing Value Treatment**
  In **text columns**, missing values are replaced with "NA".
  In **whole number columns**, missing values are replaced with 0.
- **Bag Weight Splitting**
  The **Bag Weight** column is split into two separate columns: one containing the numeric value and the other containing the unit (kg), using Power Query Editor.
- **Numerical Data Trimming**
  Leading and trailing spaces, as well as unwanted characters, are removed from numeric fields in Power Query Editor.

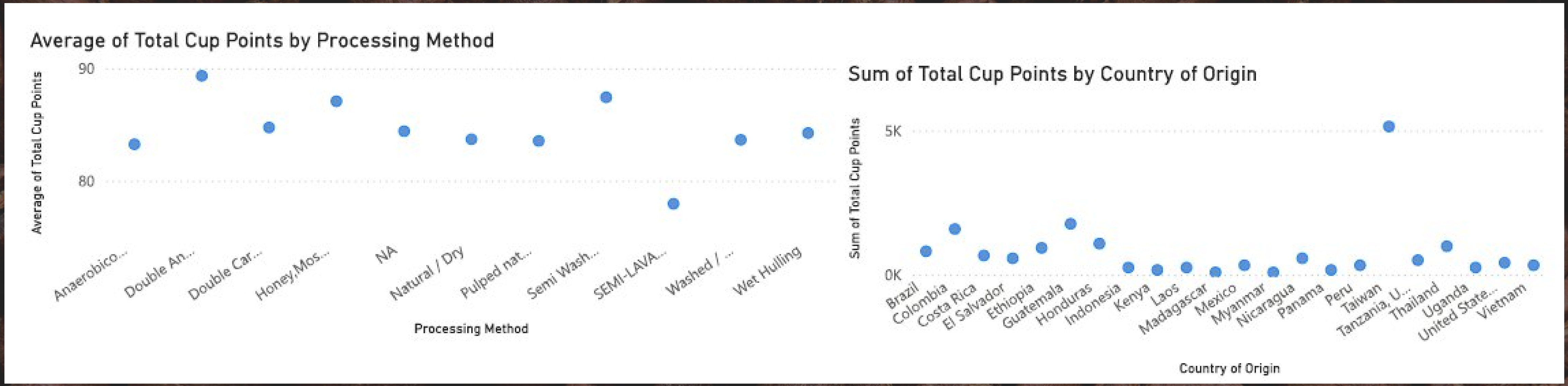
DATA VISUALIZATION AND ANALYSIS - DETERMINANTS OF COFFEE QUALITY USING DECOMPOSITION TREE VISUALIZATION

The decomposition tree shown in the visual breaks down the **total cup points** of coffee based on key sensory attributes such as **Acidity, Aftertaste, Aroma, Clean Cup, Color, and Flavor**. Each branch of the tree represents how individual attribute scores contribute to the overall coffee quality.

- **Acidity**: This is the first level of the tree, showing how variations in acidity scores influence total cup points. Higher acidity values (e.g., 7.75) correspond to higher contributions to the total score.
- **Aftertaste and Aroma**: These attributes form the next levels, further splitting the contribution of total cup points based on their respective scores. For example, a coffee sample with high acidity and a high aftertaste score contributes more to the total cup points.
- **Clean Cup, Color, and Flavor**: These final branches show how other key attributes affect the overall score. Notably, the **Clean Cup** attribute shows a maximum score of 10, indicating its significant influence. Similarly, **Color** differentiates contributions based on green vs. brown beans, reflecting quality differences.

Overall, the decomposition tree provides a clear hierarchy of attributes, allowing us to **identify which sensory characteristics have the strongest impact on total cup points**. This helps in understanding quality drivers and prioritizing improvements in coffee evaluation.



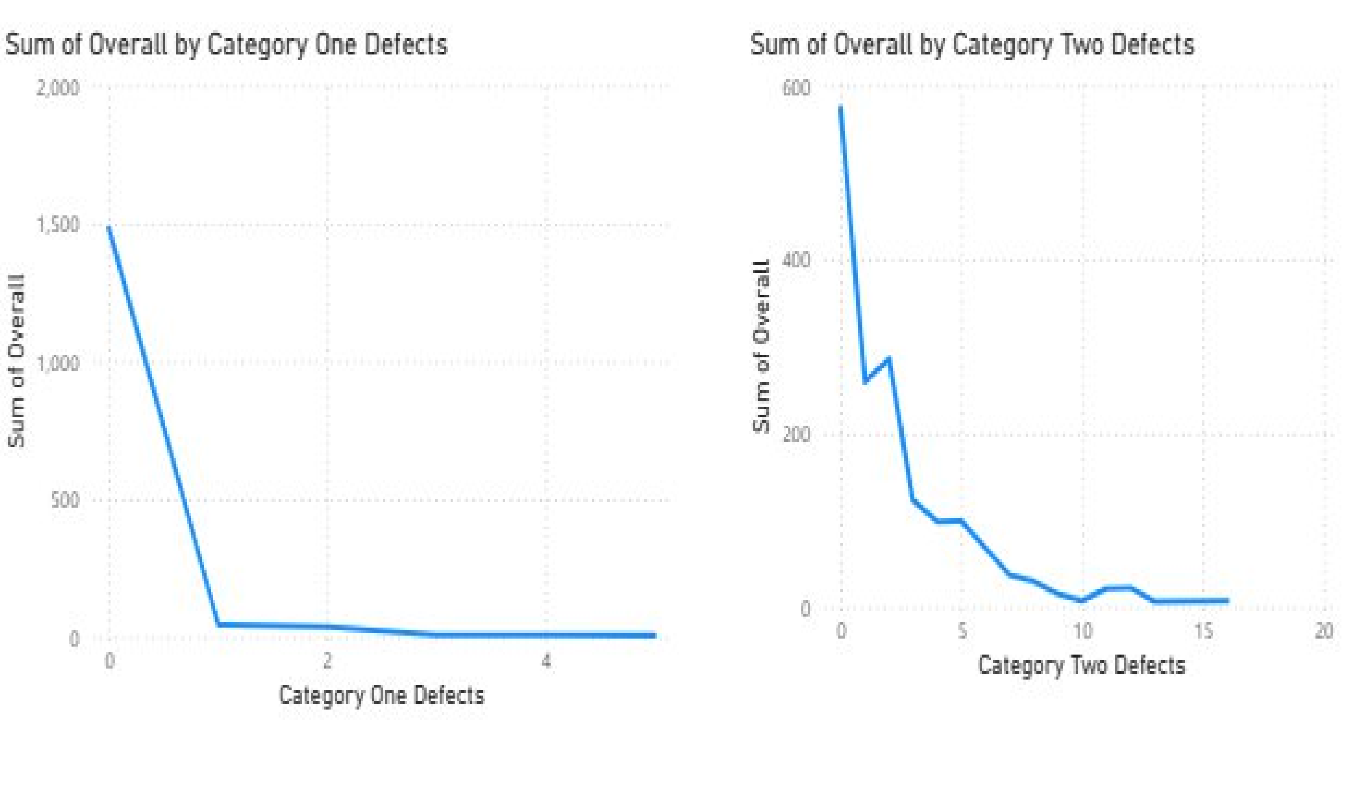


Graph 1:
•Methods like Double Carbonic Maceration and Anaerobic Washed appear to yield higher average scores, suggesting they may produce more flavorful or refined coffee.
•More common methods like Washed or Natural / Dry have slightly lower averages, but still respectable.
•The presence of "NA" (not available) indicates some data points lacked processing method info.

Graph 2:
•The data presented in graph 2 highlights the distribution of Total Cup Points across different Countries of Origin. The scatter plot provides a clear visual representation, allowing us to quickly identify patterns and anomalies in the dataset. Notably, the plot reveals an outlier: Taiwan. This outlier stands apart from the other data points, showing significantly higher Total Cup Points compared to most other countries. Such a deviation may indicate a higher demand for coffee from Taiwan, potentially due to superior quality, unique processing methods, or a niche market preference. Identifying these outliers is crucial as they provide actionable insights for market analysis, targeted sourcing strategies, and understanding trends in global coffee quality and consumption.

CORRELATION BETWEEN PROCESSING METHODS, ORIGIN REGIONS, AN D COFFEE QUALITY SCORES

TRENDS OR PATTERNS IN DEFECT OCCURRENCES AND T HEIR IMPACT ON OVERALL COFFEE QUALITY

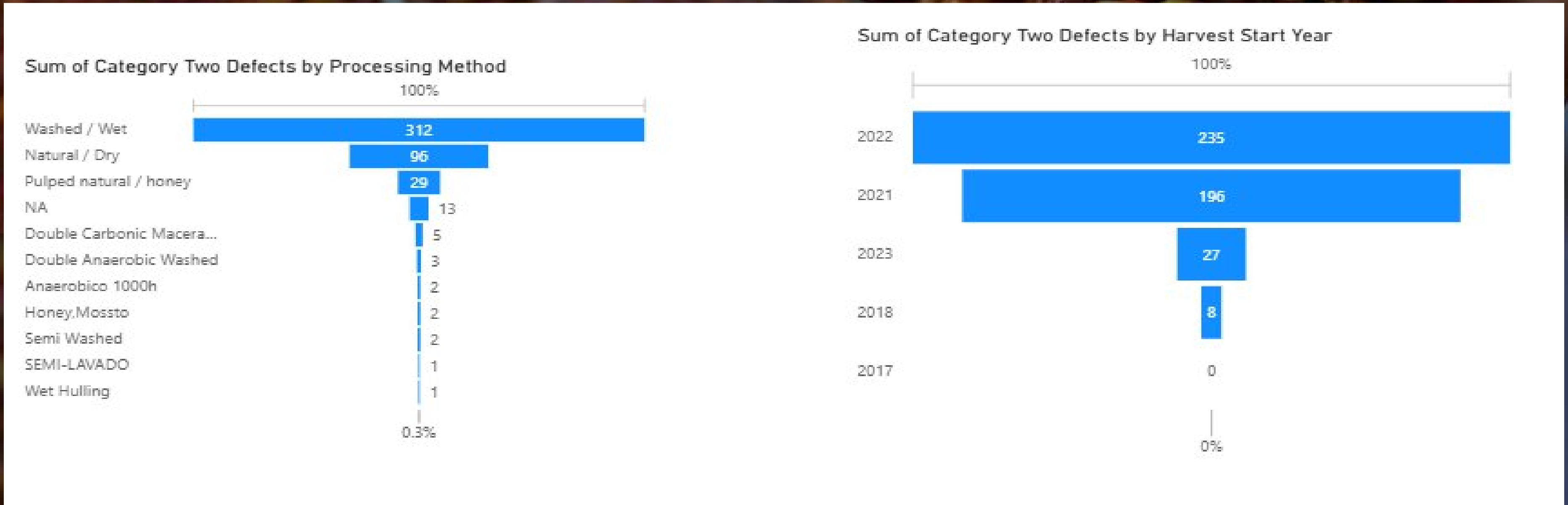


As defects increase → Score drops steadily, but the decline is less steep than Category One defects.

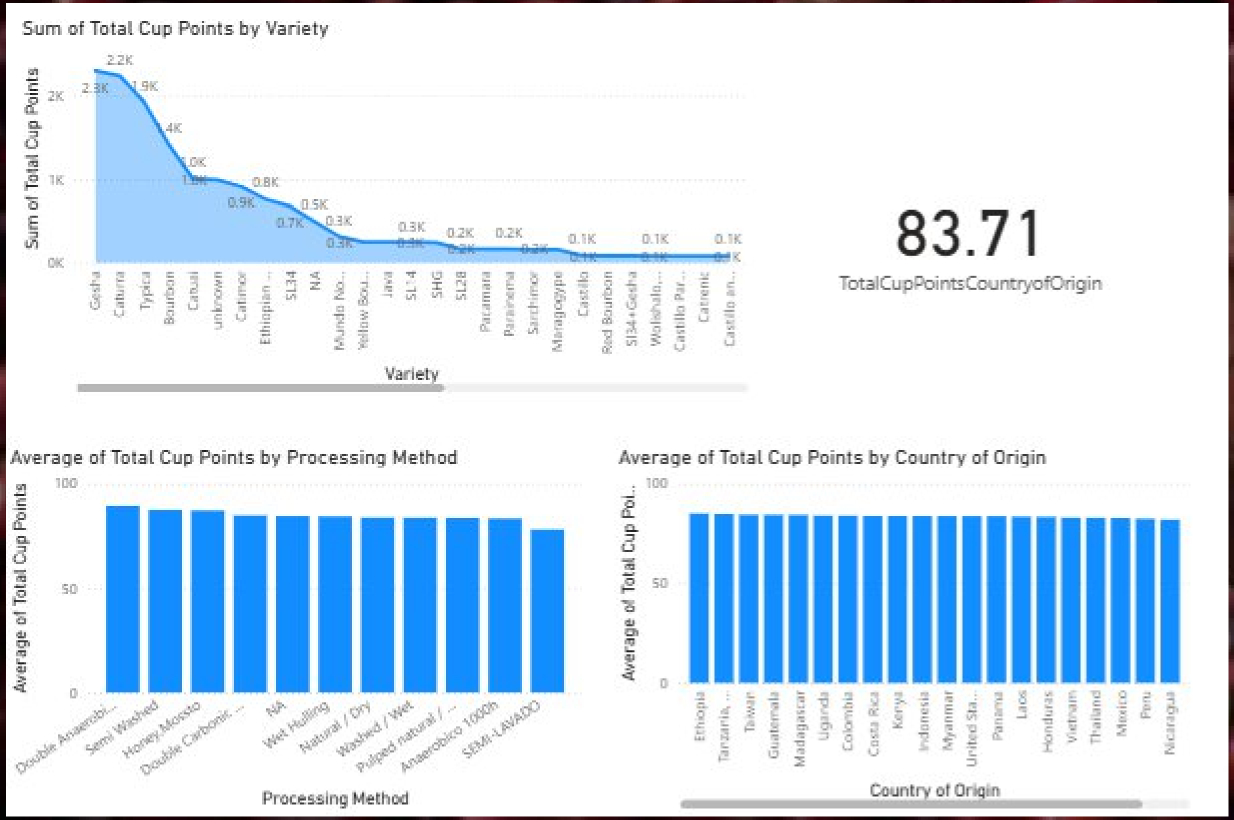
- Category One defects are fewer but more critical — even 1–2 defects drastically reduce coffee quality.
- Category Two defects occur more frequently but each defect has a smaller individual impact.
- The correlations support this: Category Two defects have a stronger overall correlation (0.81), indicating they collectively affect total quality more, while Category One defects are very impactful individually but may be fewer in occurrence.

CORRELATION BETWEEN PROCESSING METHOD AND HARVEST START YEAR WITH CATEGORY TWO DEFECT

The analysis of defects by processing method and harvest year reveals that the Washed and Wet processing methods have the highest number of Category 2 defects. This indicates that coffees processed using these methods are more prone to intermediate-level quality issues, which could be due to specific handling, fermentation, or drying practices inherent to these methods. Additionally, the data shows that the harvest year 2022 recorded the highest number of Category 2 defects across all samples. This trend may reflect environmental factors, changes in harvesting techniques, or other seasonal variables that impacted coffee quality during that year. Understanding these patterns is essential for quality control, as it allows producers and buyers to pinpoint potential areas of improvement and better manage risks associated with defects.



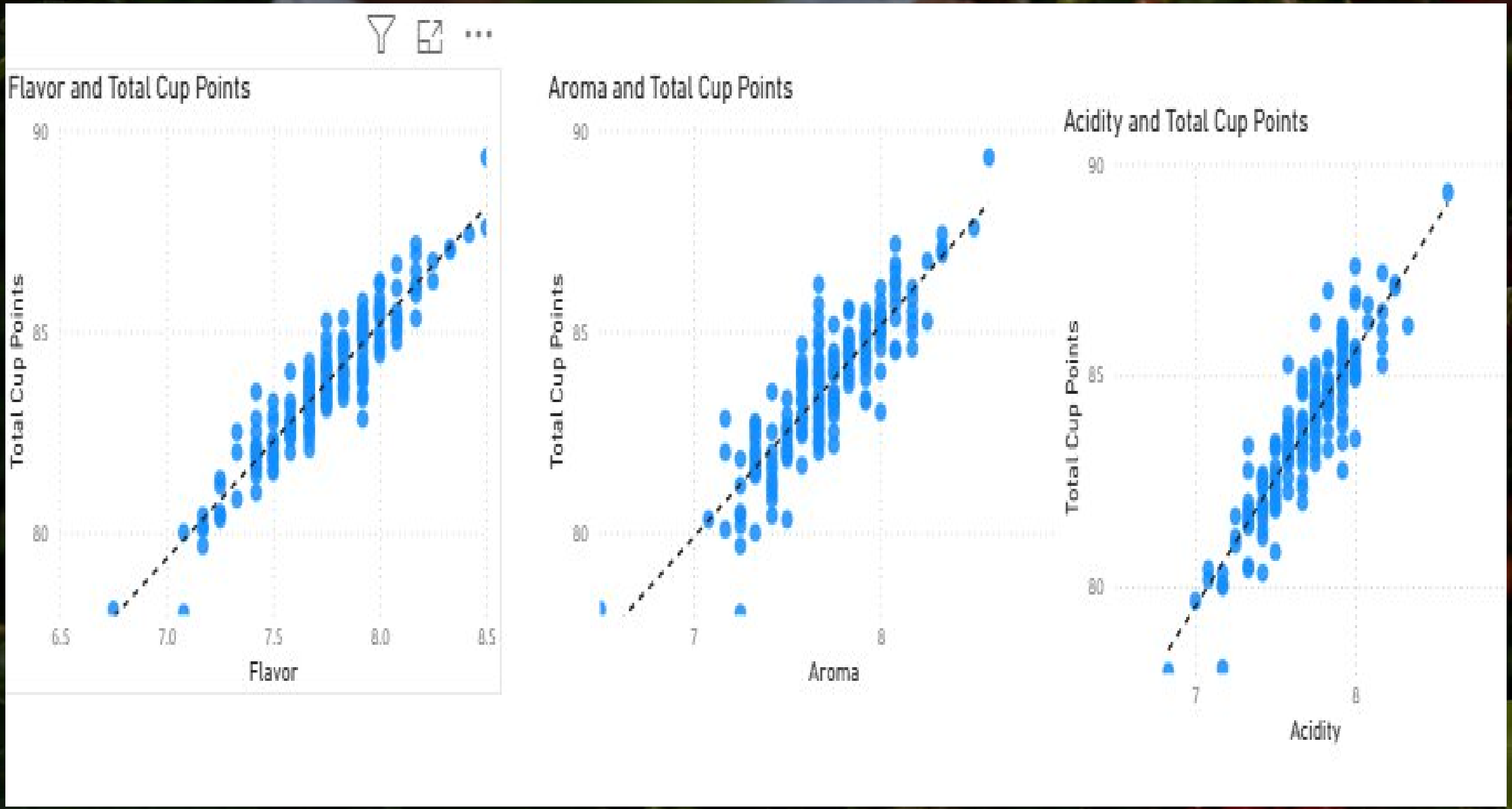
DIFFERENT VARIABLES INTERACT TO INFLUENCE THE TOTAL CUP POINTS, WHICH REPRESENT AN OVERALL MEASURE OF COFFEE QUALITY



The bar charts and line curve shows how different variable interacts with Total cup points. The sum of Total Cup Quality is highest for Gesha variety of coffee that means its demand is more and Castillo and Colombia blend has the lowest total cup points. In processing method, the average total cup score od Double anaerobic washed processing method is highest that indicate more consumption. Country-wise Ethiopia has the highest total cup points.

DIFFERENT VARIABLES INTERACT TO INFLUENCE THE TOTAL CUP POINTS, WHICH REPRESENT AN OVERALL MEASURE OF COFFEE QUALITY - EXTENDED VERSION

In below scattered plot shows flavour, acidity and aroma are directly related to coffee quality.



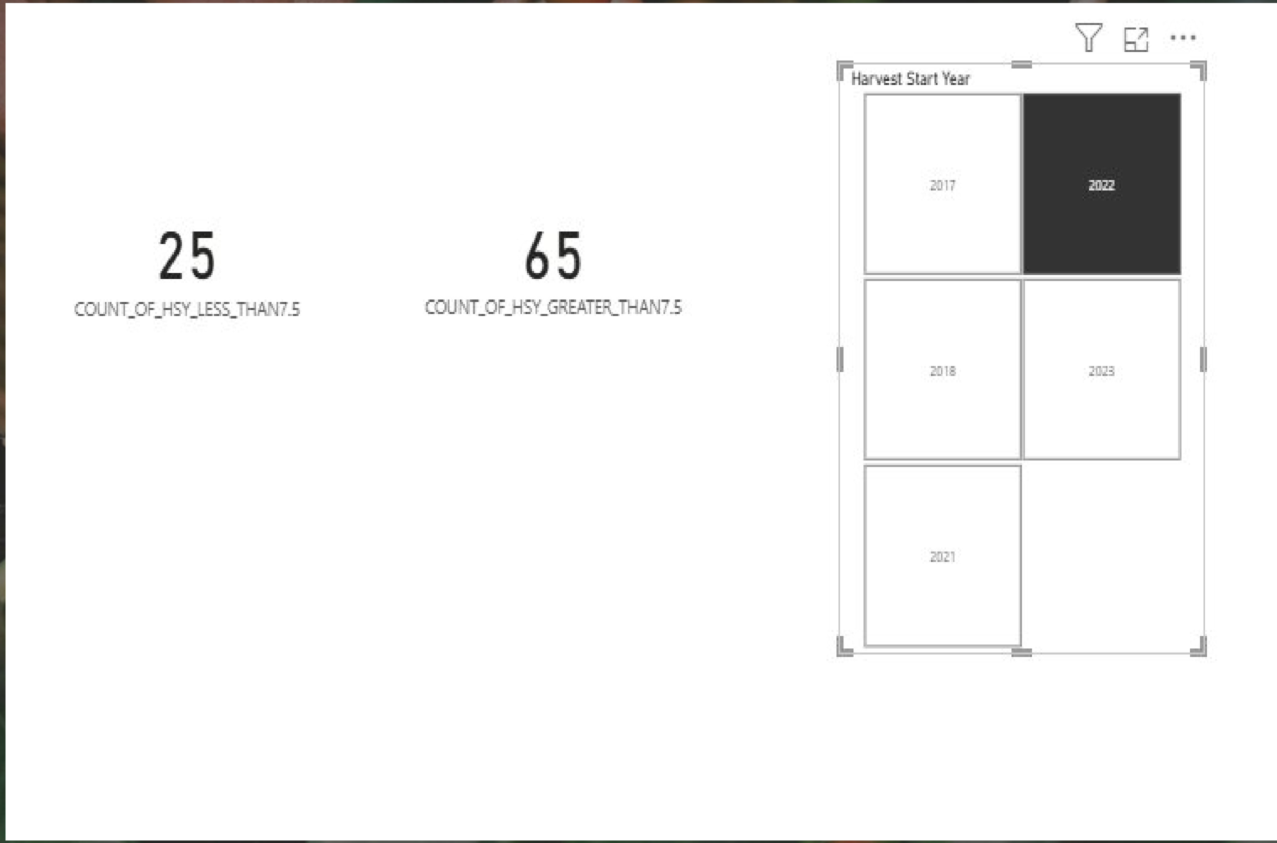
COUNTRY AND REGION PRODUCE RICH QUALITY OF COFFEE(RICHNESS OF COF FEE=OVERALL>7.5)

The conditional analysis focuses on coffee samples with a richness score of 7.5. A table visualization is used to provide a clear and organized view of which coffee IDs, countries of origin, and regions fall into this richness category based on their overall cup points. This approach allows us to quickly identify high-quality coffees and observe patterns related to geographic origin or specific regions. By segmenting the data in this way, we can better understand how richness correlates with other sensory attributes and pinpoint which origins consistently produce coffees with superior

ID	Overall	GreaterinRichness	Region	Country of Origin
122	7.50	True	NEW ORIENTE & HUEHUETENANGO	Guatemala
126	7.50	True	Yunlin	Taiwan
128	7.50	True	KILIMANJARO	Tanzania, United Rep
131	7.50	True	Quindio	Colombia
132	7.50	True	Marcala, La Paz	Honduras
134	7.50	True	Hsinchu County	Taiwan
138	7.50	True	The Plans of Santa Maria, La Paz	Honduras
141	7.50	True	ZONGOLICA, VERACRUZ	Mexico
144	7.50	True	mark	Honduras
149	7.50	True	Wheels	Colombia
152	7.50	True	the plans of Santa Maria la Paz	Honduras
153	7.50	True	Miaoli County	Taiwan
154	7.50	True	Miaoli County	Taiwan
159	7.50	True	Kona	United States (Hawai
161	7.50	True	Center, Lagunetillas - Ajuterique, Comayagua	Honduras
170	7.50	True	Kericho	Kenya
172	7.50	True	Taoyuan City	Taiwan
173	7.50	True	Miaoli County	Taiwan
176	7.50	True	Corralillo Tarrazu	Costa Rica
177	7.50	True	East	Guatemala
101	7.58	True	Cauca	Colombia
103	7.58	True	tolima	Colombia
107	7.58	True	South Shan State	Myanmar
114	7.58	True	Popayán Cauca	Colombia
115	7.58	True	North of Thailand	Thailand
124	7.58	True	Sierra Nevada de Santa Marta	Colombia
130	7.58	True	Miaoli County	Taiwan
136	7.58	True	Nongluang Bolaven Plateau, Champasack, Lao PDR	Laos
137	7.58	True	Center, Lagunetillas - Ajuterique, Comayagua	Honduras

RELATION BETWEEN HARVESTING START YEAR AND RICHNESS

The year-wise analysis has been conducted by categorizing coffee samples based on their richness scores. Observing the data across different harvest years, it is evident that coffees harvested in 2022 and 2023 exhibit higher richness compared to other years. This indicates a trend of improving quality in terms of richness over recent harvests, which could be influenced by factors such as favorable climatic conditions, better cultivation practices, or advancements in post-harvest processing. Such insights help in understanding temporal patterns in coffee quality and can guide decisions related to sourcing and quality assessment.



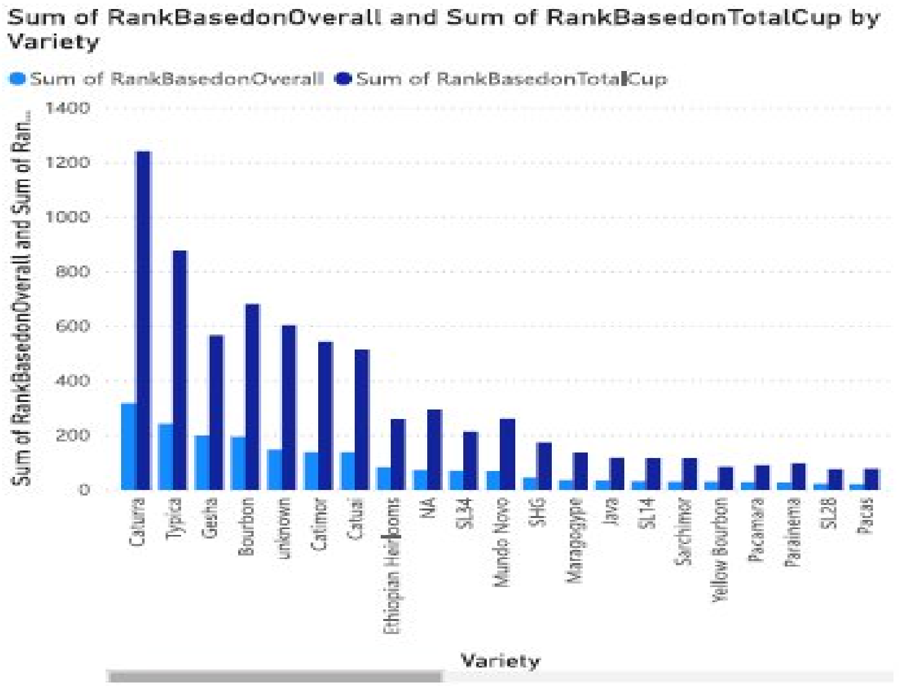
CONDITIONAL STATEMENT

Lot Number	Total weight	Below1000kg
010/0296/600	300.00	True
01-03-2022	20.00	True
0423A01	2.00	True
0523B01	2.00	True
0623C01	2.00	True
1	2.00	True
1	300.00	True
2021/01	50.00	True
2021/11	20.00	True
202112	8.00	True
2022 Tainan City Specialty Coffee Evaluation Batch	20.00	True
2022 Tainan City Specialty Coffee Evaluation Batch	80.00	True
2022 Tainan City Specialty Coffee Evaluation Batch	100.00	True
2022/01	10.00	True
2022/02	150.00	True
2022/3/1(Lao Cong)	144.00	True
202203	10.00	True
202203	20.00	True
202203	100.00	True
202203	300.00	True
21/22	300.00	True
221221	1.00	True
A01-Northern Taiwan Specialty Coffee Evaluation	4.00	True
A02-Northern Taiwan Specialty Coffee Evaluation	200.00	True
A03-Northern Taiwan Specialty Coffee Evaluation	13.00	True
A04-Northern Taiwan Specialty Coffee Evaluation	8.00	True
A05-Northern Taiwan Specialty Coffee Evaluation	13.00	True
A10-Northern Taiwan Specialty Coffee Evaluation	6.00	True

In the conditional statement, the table indicated that the lot numbers are weightage below 1000kg and there are total 112 lot numbers below 100kg. Allowing buyers to accept coffee lots below 1000 kg would involve a large portion of the market's inventory (112 lots in this case). This could benefit both buyers seeking smaller amounts and sellers with limited batch sizes, broadening market participation and potentially increasing sales for smaller producers.

DISPLAY THE RANK OF ID, LOT NUMBER BASED ON OVERALL AND TOTAL CUP

ID	Lot Number	Region
0	CQU2022015	Piendamo,Cauca
1	The 2022 Pacific Rim Coffee Summit,T037	Chiayi
2	The 2022 Pacific Rim Coffee Summit,LA01	Laos Borofen Plateau
3	CQU2022017	Los Santos,Tarrazu
4	CQU2023002	Popayan,Cauca
5	The 2022 Pacific Rim Coffee Summit,GT02	Chimaltenango
6	The 2022 Pacific Rim Coffee Summit,T034	Chiayi
7	The 2022 Pacific Rim Coffee Summit,T050	Chiayi
8	The 2022 Pacific Rim Coffee Summit,T018	Chiayi
9	CN 4127230034/4189230113	KILIMANJARO
10	010/0296/600	Guji
11	The 2022 Pacific Rim Coffee Summit,GT12	Acatenango
12	The 2022 Pacific Rim Coffee Summit,T051	Yunlin
13	Grade 1, Guji, Natural, Gelana Geisha	Guji
14	The 2022 Pacific Rim Coffee Summit,CO03	tolima
15	The 2022 Pacific Rim Coffee Summit,T014	Chiayi
16	CQU2023006	Gedeb,Yirgacheffe,Sidamo
17	202203	Shibi, Gukeng Township, Yunlin County Postal Code, Taiwa
18	202112	Gukeng Township, Yunlin County
19	CN 4127230032/P-9140	Arusha
20	The 2022 Pacific Rim Coffee Summit,GT03	Guatemala, Fraijanes, Santa Rosa
21	01-03-2022	Zhuoxi Township
22	1	Chiang Mai
23	CQU2023005	Quindio



From the table, it is shown that ID number 0 and lot number CQU2022015 has the highest rank in terms of overall and total cup, the varieties "caturra" and "typica" appear to have the highest sums in both ranking categories. Other varieties like bourbon, catuai, geisha, anaerobic, and others show smaller sums of rankings.

SUMMARY

At the end, it can be said that This project focuses on analysing the quality of coffee using sensory attributes and key determinants such as aroma, flavor, acidity, body, and richness, measured through overall cup points. The analysis combines data visualization, conditional analysis, and statistical insights to understand patterns in coffee quality across different countries, regions, processing methods, and harvest years.

Key highlights include:

Determinants of Quality: Sensory attributes like aroma, flavour, acidity, body, and richness are critical in evaluating overall coffee quality.

Outlier Identification: Scatter plots revealed outliers, such as Taiwan, indicating higher total cup points compared to other origins.

Processing Methods: Washed and wet processing methods were observed to have higher Category 2 defects, suggesting a potential impact on quality.

Harvest Year Trends: Year-wise analysis showed that coffees from 2017 and 2022 had higher richness scores, reflecting improved quality over recent years.

Conditional Analysis: Using a richness threshold (e.g., 7.5), the project identified which coffee IDs, countries, and regions consistently fall into high-quality categories.

Ranking Analysis: The analysis ranks coffee lot numbers by overall and total cup scores, showing that Caturra and Typica varieties hold the highest ranks, indicating top quality. Other notable varieties include Bourbon, Catuai, and Geisha with moderate ranks. Samples come from diverse coffee regions. This indicates a focus on quality assessment in that event and highlights leading coffee varieties in terms of overall and cup performance

Visual Insights: Tables, scatter plots, and decomposition trees were used to visualize relationships between origin, processing methods, defects, and sensory scores, enabling easier identification of patterns and anomalies.

The project provides actionable insights into factors affecting coffee quality and highlights trends across regions and years, aiding stakeholders in quality assessment, sourcing decisions, and process improvement.

THANK YOU

Email
sbhattacherjee98@gmail.com